

Chapter 13

Volume

Unit 4 — Area and Volume

McGraw Hill Glencoe Geometry Texas, 2015

Key Vocabulary

Term	Definition
Volume	Amount of space enclosed by a solid (cubic units)
Prism	Two congruent parallel bases; lateral faces are parallelograms
Cylinder	Two congruent circular bases
Pyramid	One base; triangular lateral faces meeting at apex
Cone	One circular base; comes to an apex
Sphere	All points equidistant from center
Composite solid	Combination of two or more solids

Cavalieri's Principle

If two solids have the **same height** and **equal cross-sectional areas** at every level, they have the **same volume**.

This is why the volume formulas for oblique and right solids are the same.

Theorem 13-1: Volume of a Prism

$$V = Bh$$

where B = area of base, h = height.

Rectangular prism: $V = lwh$

Triangular prism: $V = \frac{1}{2}bh_{\triangle} \cdot h_{\text{prism}}$

Example — Rectangular Prism

$$l = 8, w = 5, h = 3$$

$$V = 8 \times 5 \times 3 = 120 \text{ cubic units}$$

Theorem 13-2: Volume of a Cylinder

$$V = \pi r^2 h$$

Example: Radius = 4, height = 9:

$$V = \pi(4)^2(9) = 144\pi \approx 452.4$$

Theorem 13-3: Volume of a Pyramid

$$V = \frac{1}{3}Bh$$

where B = area of base, h = height.

Note: Volume of pyramid = $\frac{1}{3} \times$ volume of prism with same base and height.

Example — Square Pyramid

Base: 6×6 , height = 10

$$B = 36, \quad V = \frac{1}{3}(36)(10) = 120$$

Theorem 13-4: Volume of a Cone

$$V = \frac{1}{3}\pi r^2 h$$

Note: Volume of cone = $\frac{1}{3} \times$ volume of cylinder with same radius and height.

Example: Radius = 3, height = 7:

$$V = \frac{1}{3}\pi(9)(7) = 21\pi \approx 66.0$$

Theorem 13-5: Volume of a Sphere

$$V = \frac{4}{3}\pi r^3$$

Example: Radius = 6:

$$V = \frac{4}{3}\pi(216) = 288\pi \approx 904.8$$

Volume Summary

Solid	Formula
Prism	$V = Bh$
Cylinder	$V = \pi r^2 h$
Pyramid	$V = \frac{1}{3}Bh$
Cone	$V = \frac{1}{3}\pi r^2 h$
Sphere	$V = \frac{4}{3}\pi r^3$

Memory tip: Pyramid/Cone = $\frac{1}{3} \times$ corresponding Prism/Cylinder

Surface Area vs. Volume

Solid	SA	Volume
Sphere	$4\pi r^2$	$\frac{4}{3}\pi r^3$
Cylinder	$2\pi rh + 2\pi r^2$	$\pi r^2 h$
Cone	$\pi r\ell + \pi r^2$	$\frac{1}{3}\pi r^2 h$

Example — Composite Solid

A cylinder (radius 4, height 6) is topped with a cone (radius 4, height 3).

$$V_{\text{cylinder}} = \pi(16)(6) = 96\pi$$

$$V_{\text{cone}} = \frac{1}{3}\pi(16)(3) = 16\pi$$

$$V_{\text{total}} = 112\pi \approx 351.9$$

Similar Solids

If two similar solids have scale factor k :

Measurement	Ratio
Linear (edge, radius, height)	k
Area (SA, base area)	k^2
Volume	k^3

Example: Scale factor $\frac{2}{3}$:

- SA ratio = $\frac{4}{9}$
- Volume ratio = $\frac{8}{27}$

Chapter 13 — Formula Summary

Solid	Volume
Rectangular Prism	lwh
General Prism	Bh
Cylinder	$\pi r^2 h$
Pyramid	$\frac{1}{3} Bh$
Cone	$\frac{1}{3} \pi r^2 h$
Sphere	$\frac{4}{3} \pi r^3$

Similar Solids Scale k	Ratio
Linear	k
Area	k^2
Volume	k^3